EXPERIMENT - 3 (Study)

EXP NO:3

STUDY OF NETWORK CABLES AND CRIMPING WITH RJ45 CONNECTOR

AIM:

To study different types of network cables & to prepare a straight through & cross-over Ethernet cable by crimping RJ45 connectors

THEORY:

1. Twisted pair cable:

UTP (unshilded) & STP (Shielded). used in LANS

categories:

→ cat5
→ cat5e
→ cat6
→ cat6a

2. coaxial cable.

used in cable TV & other ethernet (10Base2 & 10Base5)

Higher shelding

3. Fiber optic cable:

Transmits data as light pulses

Single mode (long distance)

multimode (short distance)

Very high bandwidth

Wiring Standards (Pinout)								
TIA/EIA 568B (most common)								
Pin No.	1	2	3	4	5	6	7	8
Color	Orange-White	Orange	Green-White	Blue	Blue-White	Green	Brown-White	Brown
TIA/EIA 568A								
Pin No.	1	2	3	4	5	6	7	8
Color	Green-White	Green	Orange-White	Blue	Blue-White	Orange	Brown-White	Brown

Straight-Through: 568B at both ends (PC ↔ Switch)
Cross-Over: 568B at one end and 568A at the other (PC ↔ PC)

→ RJ45 is the standard connector for Ethernet cables

Two wiring standard exist

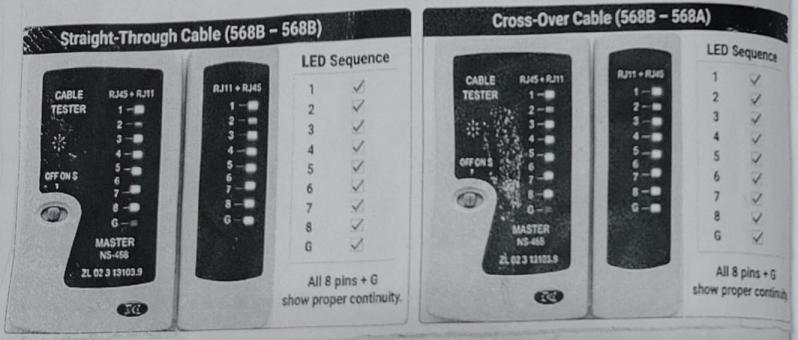
→ TIA/EIA 568A:

Green - white, Green, orange - white Blue, blue - white, orange, Brown - white, Brown

- TIA/EIA 568B.

orange - white, orange, Green - White, Blue, Blue - White, Green, Brown - White, Brown

Cable Tester Results



RESULT:

Both straight-through & cross-over Ethernet cables were successfully crimped & verified using a cable testers. All 8 pins showed proper continuity